Fuzzy–Neural Decision Intelligence for Cost-Aware Cyber Alert Triage

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Abstract

Security operations centres often receive strong alert scores from machine learning models, but the step from score to operational action is still handled with brittle thresholds. This short communication proposes a fuzzy–neural decision intelligence framework that treats cyber alert triage as a knowledge-based decision problem rather than a classification endpoint. A compact multilayer perceptron estimates attack propensity from NSL-KDD features. Triangular fuzzy membership functions convert propensity into linguistic Low / Medium / High risk bands. A cost-aware decision layer maps those bands to Monitor, Soft-block, or Escalate actions under an explicit relative cost model. In a reproducible CPU-scale experiment, gradient boosting attains the highest ROC-AUC (0.947). The fuzzy–neural policy, using the neural propensity scores, reduces mean expected triage cost to 3.069 versus 3.264 for a hard 0.5 threshold and 4.309 for a top-20% escalate rule, while raising attack coverage to 0.686. The contribution is decision-facing knowledge representation and cost-aware actioning, not a new state-of-the-art detector.

Keywords: fuzzy systems, neural networks, decision intelligence, intrusion detection, intelligent decision support, knowledge-based systems

1. Introduction

Artificial intelligence and knowledge-based methods already support prediction in security operations. For intrusion alerts, analysts still need a sec-

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ond answer beyond accuracy: which events to monitor, soft-contain, or escalate under limited response capacity.

Accuracy-centric detectors optimise ROC-AUC or F1 and stop at a score or label. Pure expert rule bases are transparent but brittle when traffic shifts. Fuzzy–neural hybrids learn from data while exposing linguistic grades operators can inspect, matching the aims of *Knowledge-Based Systems* for intelligent decision support and computational intelligence.

Research gap.. Prior fuzzy and neural IDS work mainly lifts detection rates. Fewer studies close the loop from learned propensity to linguistic knowledge bands to cost-aware triage actions, and then evaluate that loop with decision metrics rather than classifier metrics alone.

Contributions.. (1) A three-layer fuzzy–neural decision pipeline for cyber alert triage. (2) An evaluation protocol reporting decision quality alongside predictive metrics on public NSL-KDD data. (3) A reproducible short-cycle CPU experiment with open artifacts.

2. Related work

Machine learning for intrusion detection on NSL-KDD and related corpora is extensively surveyed [1, 2, 3]. Fuzzy and neuro-fuzzy IDS designs improve detection and interpretability [4, 5, 6, 7]. Cost-sensitive learning formalises asymmetric errors [8]. Security analytics emphasises prioritisation and incident prediction [9, 10]. Explainability methods such as SHAP can aid trust [11], but explanation alone does not select Monitor / Soft-block / Escalate under budget. We claim a compact knowledge-based decision layer evaluated with cost and coverage, not a new highest AUC.

3. Methodology

3.1. Problem formulation

For features x_i and label $y_i \in \{0, 1\}$, the system outputs propensity $p_i = f_\theta(x_i) \in [0, 1]$, fuzzy memberships μ_L, μ_M, μ_H, and action $a_i \in \{\text{Monitor}, \text{SoftBlock}, \text{Escalate}\}$.

3.2. Neural propensity model

An MLP (128–64–1) with ReLU, dropout, and class-weighted BCE estimates p_i. Numeric features are standardised; protocol, service, and flag are one-hot encoded [12].

Table 1: Predictive performance on NSL-KDD test sample (seed 42).

Model	Acc.	F1	ROC-AUC	PR-AUC
Logistic regression	0.752	0.740	0.818	0.879
Gradient boosting	0.786	0.774	**0.947**	**0.951**
Neural net (ablation)	0.773	0.766	0.908	0.923
Fuzzy–neural (proposed)	0.773	0.766	0.908	0.923

Table 2: Decision quality using neural propensity scores.

Policy	Mean cost	Attack coverage	Unnec. escalations
Prob. threshold 0.5	3.264	0.655	623
Top-20% by score	4.309	0.348	47
Fuzzy–cost (proposed)	**3.069**	**0.686**	603

3.3. Fuzzy risk banding

Triangular memberships on p define Low / Medium / High risk [13]. Dominant membership yields an inspectable linguistic band.

3.4. Cost-aware decision layer

Relative costs penalise missed attacks and unnecessary escalations. Dominant Low→Monitor, Medium→SoftBlock, High→Escalate. Baselines: hard threshold 0.5; escalate top 20% by score.

4. Experimental setup

NSL-KDD train/test [3], binary normal vs attack. Training subsampled to 40,000 rows; test capped at 20,000 (seed 42). Baselines: logistic regression; gradient boosting [14]; neural ablation; proposed fuzzy–neural policy. Metrics: Accuracy, F1, ROC-AUC, PR-AUC; mean expected cost; attack coverage; unnecessary escalations. CPU-only Python/PyTorch run well under two hours.

5. Results

Table 1 shows gradient boosting leads on AUC. Proposed prediction metrics match the neural ablation by design. Table 2 and Fig. 2 show the fuzzy–

Fig. 1. Fuzzy–neural decision intelligence pipeline for cyber alert triage

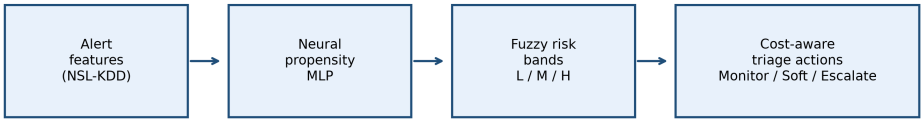


Figure 1: Fuzzy–neural decision intelligence pipeline for cyber alert triage.

Fig. 2. Predictive ROC and cost-aware triage comparison

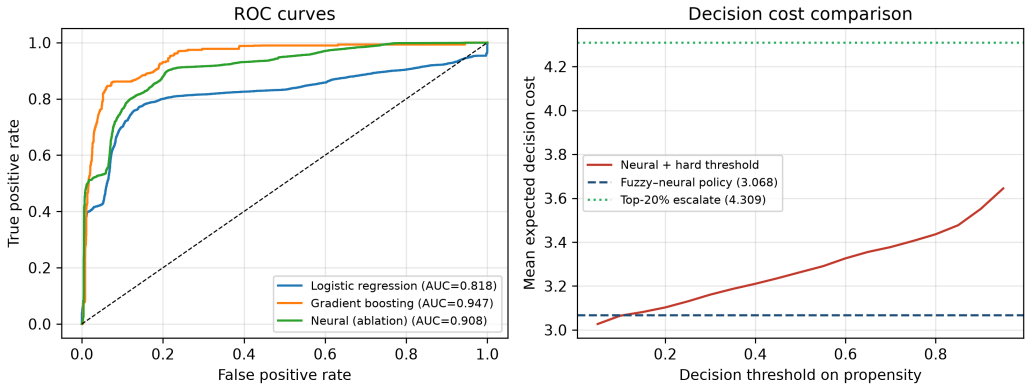


Figure 2: ROC curves and mean expected decision cost versus threshold.

neural policy improves mean triage cost and attack coverage versus simple score-to-action baselines, while Fig. 1 summarises the knowledge-facing pipeline.

6. Discussion

The design combines learning, fuzzy knowledge presentation, and decision support for prediction/warning settings. We do not claim state-of-the-art NSL-KDD detection; pairing the fuzzy–cost layer with stronger propensity models is a natural extension. Limitations include public benchmark traffic, scenario-default costs, and training subsample for speed.

7. Conclusion

We presented a fuzzy–neural decision intelligence framework for cost-aware cyber alert triage. Reproducible NSL-KDD results show improved decision cost and coverage under linguistic Monitor / Soft-block / Escalate actions. Future work includes ANFIS-style adaptive memberships, operator-calibrated costs, and phishing-transfer evaluation.

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Data and code availability

Experiment code, tables, and figures: `https://github.com/nithin427/fuzzy-neural-cyber-triage`.

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